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Develop a Closed-Loop Stepper Motor Control System with UI for Software Configuration

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CHAPTER I

INTRODUCTION

This chapter established the foundation of the study by presenting the overall structure of the research on developing a closed-loop stepper motor system with a user interface for software configuration. It shows the research objectives that guided the design and development of the locally made controller, implementing a manual configuration for parameter adjustment, and creating a desktop user interface for device monitoring and control. It also described the conceptual framework that guided the process for both hardware and software and clarified the scope and limitations of the study to show what the system can do and what it does not do. Finally, key terms related to stepper motor control, closed-loop operation, and graphical user interfaces were defined to help readers understand the important concepts used throughout the study.

1.1 BACKGROUND OF THE STUDY

Stepper motors are widely used around the world, especially in automation and precision control system. The researchers look at the data on where the stepper motor is usually used and they can be found in many places, such as factories, laboratories, 3D printers, and robotic platforms, it is because the stepper motor moves in discrete steps. One study on stepper motor control noted that, without closed loop feedback, these motors can suffer from missed steps and positioning

errors, especially under varying loads and high speeds (DrPress, 2022). This problem is common on low-cost systems, where the motor just follows what the user inputs a command without verifying if the motor actually rotates or moves that much, which can lead to step loss, stalling, and reduced accuracy.

The researchers have looked a different ways on how to reduced these problems. Some studies have introduced closed-loop control that uses a sensors such as encoders or current based feedback to detect position error. Other studies used advanced algorithms for better handling of loads and speed and they also tried to create a PC based or touch screen interface to allow users to adjust the parameters for making motor control more accessible to students.

Because of this, the researchers was inspired to develop a closed-loop stepper motor control system that is locally implementable and easy to use. The study aims to develop a locally made controller, and desktop user interface for software integration. The system will allow the user to set motion parameters, and control movement through the user interface, making advanced motion control be easily use and cost-effective for students and small scale projects.

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1.2 OBJECTIVES OF THE STUDY

The primary objective of the study was to develop a controller for a closed-stepper motor system locally with a user interface. This user interface was designed to navigate the system's functions and controls, to guide the users on how the system works while ensuring the operation of the controller was intuitive and easy to use.

To elaborate further, the study aimed to:

1. Construct a controller locally for a closed-loop stepper motor and implement a manual configuration for parameter adjustment
2. Develop a desktop executable user interface for device monitoring and controls
3. Conduct various testing procedures to evaluate the performance of the system to ensure compliance with criteria and standards.

1.3 SCOPE AND LIMITATIONS

This study focuses on the development of a closed-loop stepper motor control system with a UI for software configuration, which aims to design and implement a locally made controller to address common problems such as step loss, stalling, and inaccurate positioning. The system includes manual configuration so users can adjust the parameters and it also includes a desktop user-interface for monitoring the motor, set parameters and control motions without reprogramming the systems. Finally, the system will be conducting a various testing procedures to evaluate the overall performance of the system.

The study has several limitations,The study is developed and tested only in a controlled environment, which may not fully replicate actual industrial conditions. The system relies on a simple feedback method, it cannot completely fix inaccuracies that are caused by electrical interference or a loose motor's gear. Hardware limitations, such as power capacity, component tolerances, and communication llatency between the controller and the computer. The desktop user interface is designed only for a basic monitoring and configuration tasks and does not support any advanced features like multi-axis motion or integration with complex automation systems.

1.4 SIGNIFICANCE OF THE STUDY

This study was designed to address the need for an accessible and user-friendly closed-loop stepper motor control system suitable for academic laboratories and small-scale projects, allowing them to perform their experiments independently and consistently. Some institutions still keep relying on basic open-loop stepper motor drivers, which are more prone to step loss, inaccurate positioning, and unstable performance under heavy load. The researchers developed this study to respond to the market demand for better control systems that are easy to understand, configure, and maintain. The use of a closed-loop stepper motor control system is also designed with a desktop user interface for user-friendly access. For the tech industries and developers, they can use the system to fabricate automation experiments instead of relying on expensive imported hardware. Since the controller and software can be modified for different applications. Additionally, this research is a simple but functional system that can serve as a foundation for more advanced future studies. Future researchers could

build upon this work by integrating more advanced control algorithms, automatic calibration, and data logging for predictive maintenance, and lastly develop a mobile interface so the system can be controlled and monitored remotely.

1.5 CONCEPTUAL FRAMEWORK

The study aimed to assess the functionality and relevance of the idea of a closed-loop stepper motor control system with a desktop user interface for software configuration that has the ability to achieve better performance and user-friendly systems. This framework has several key components for conducting the study. The following system flow diagram illustrates the proposed system:

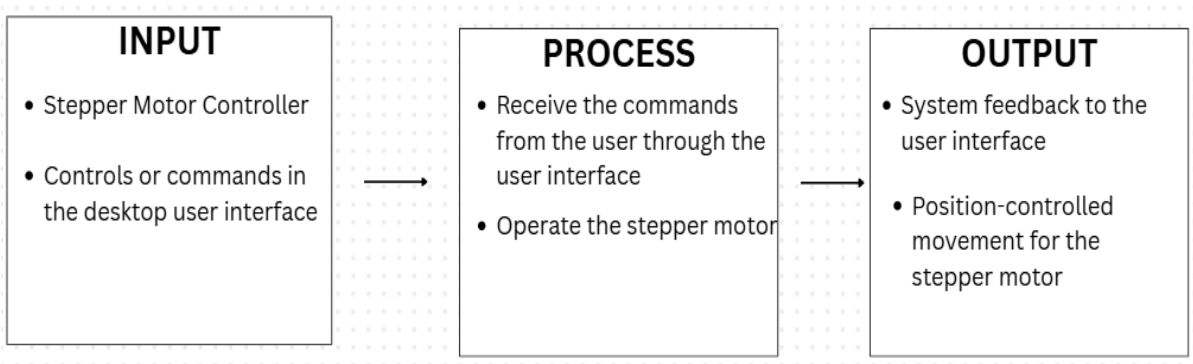

Fig.1 IPO table

Fig. 1 illustrates the process of the research paradigm, highlighting the key steps in developing the prototype. The process began with the user inputting their desired commands of the parameters for the stepper motor, followed by the generation of controlled signals to operate the stepper motor. The system was designed for controlling the stepper motor

movement with the use of a desktop user interface. The output, the system sends feedback signals to the user interface so that users can determine if there are any errors in the system.

The IPO (Input-Process-Output) framework was necessary for this study, providing a structured approach to understand the process involved in developing and designing the closed-loop stepper motor system with UI for software configuration. Inputs, such as the locally made controller for the closed-loop stepper motor and the controls in the desktop user interface, where users can input the commands, parameter updates, and monitoring requests. This method ensured that the system worked properly and delivered the desired results of the system.

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